23A01604b • OPEN CHANNEL FLOW

UNIT – 1

Hydraulics of Pipelines and Pipe Networks

Complete Study Notes • Concepts · Derivations · Formulae · Exam Questions

Syllabus Topics

- Review of Fluid Mechanics
- Reynolds Transport Theorem and Applications
- Steady Flow Analysis of Pipe Network Systems
- Unsteady Flows – Basic Equations of Water Hammer
- Solution by Method of Characteristics
- Network Analysis

1. REVIEW OF FLUID MECHANICS

1.1 Fluid properties

A **fluid** is a substance that deforms continuously under any applied shear stress. The properties that matter most in hydraulics are given below.

Property	Definition / Symbol	Value for water (20°C)
Mass density	ρ = mass/volume	998 kg/m^3
Specific weight	$\gamma = \rho g$	9.79 kN/m^3
Dynamic viscosity	μ (N·s/m^2), $\tau = \mu\, du/dy$	1.0×10^{-3} Pa·s
Kinematic viscosity	$\nu = \mu/\rho$	1.0×10^{-6} m^2/s
Bulk modulus	$K = -dp/(dV/V)$	2.2 GPa
Surface tension	σ (N/m)	0.073 N/m
Vapour pressure	p_v	2.34 kPa (abs)

Newton's law of viscosity: shear stress is proportional to velocity gradient, **$\tau = \mu\,(du/dy)$**. Fluids obeying this are called **Newtonian**.

1.2 Types of flow

- **Steady / unsteady:** properties at a point do not / do change with time ($\partial/\partial t = 0$ for steady).
- **Uniform / non-uniform:** velocity does not / does change along the flow direction.
- **Laminar / turbulent:** governed by Reynolds number $Re = \rho VD/\mu = VD/\nu$. For pipes, laminar if $Re < 2000$, turbulent if $Re > 4000$. For open channels, laminar if $Re < 500$ (Re based on hydraulic radius R).
- **Compressible / incompressible:** liquids are generally treated as incompressible, except in water-hammer analysis where compressibility is the whole point.
- **Rotational / irrotational:** presence or absence of fluid-element spin.

1.3 Fundamental equations

(a) Continuity (conservation of mass). For steady incompressible flow in a stream-tube:

$$Q = A_1V_1 = A_2V_2 = \text{constant}$$

For unsteady compressible flow the differential form is $\partial\rho/\partial t + \nabla\cdot(\rho\mathbf{V}) = 0$.

(b) Energy equation (Bernoulli). Along a streamline for steady, incompressible, frictionless flow:

$$p_1/\gamma + V_1^2/2g + z_1 = p_2/\gamma + V_2^2/2g + z_2 + h_L$$

Each term has the dimension of length (metres of fluid): **pressure head**, **velocity head** and **elevation (datum) head**. The sum is the **total head H**; the sum $p/\gamma + z$ is the **piezometric head** (hydraulic grade line, HGL) and H is plotted as the **energy grade line (EGL)**. h_L is the head loss between sections 1 and 2.

(c) Momentum equation. The net external force on a control volume equals the rate of change of momentum:

$$\Sigma F = \rho Q(V_2 - V_1) \text{ (steady flow, one inlet and one outlet)}$$

Used for forces on pipe bends, nozzles, hydraulic jump and many open-channel problems.

1.4 Head loss in pipes

Major (friction) loss – Darcy–Weisbach:

$$h_f = f\,L\,V^2 / (2\,g\,D) = 8\,f\,L\,Q^2 / (\pi^2\,g\,D^5)$$

where f is the Darcy friction factor. For laminar flow f = 64/Re. For turbulent flow f depends on Re and relative roughness ε/D and is obtained from the **Colebrook–White equation**:

$$1/\sqrt{f} = -2\log_{10}\left[\,\varepsilon/(3.7D) + 2.51/(Re\,\sqrt{f})\,\right]$$

The **Moody diagram** is the graphical form of this equation. Explicit approximations such as **Swamee–Jain** are used in computer programs:

$$f = 0.25 / \left[\log_{10}\left(\,\varepsilon/(3.7D) + 5.74/Re^{0.9}\,\right)\right]^2$$

Empirical formula – Hazen–Williams (water only, turbulent, SI units):

$$h_f = 10.67\,L\,Q^{1.852} / (C^{1.852}\,D^{4.87})$$

C = Hazen–Williams coefficient (≈ 140 new PVC/smooth, 100–110 old cast iron). Exponent 1.852 is why this formula is popular in network design: it is simple and explicit.

Minor losses:

$$h_m = K\,V^2 / 2g$$

Fitting	Typical K
Sudden contraction	0.2 – 0.5
Sudden expansion	$(1 - A_1/A_2)^2$
Entrance (sharp-edged)	0.5
Exit into reservoir	1.0
90° elbow	0.9
Gate valve (open)	0.2

★ **Key Point:** Minor losses are not "minor" when the pipe is short. A common check is: if L/D > 1000, neglect minor losses.

1.5 Pipes in series, parallel and branching

- **Series:** Q is the same in every pipe; total head loss = Σh_f.
- **Parallel:** head loss is the same in each branch; $Q = \Sigma Q_i$. Equivalent pipe: $Q = \Sigma\,(h_f/r_i)^{1/n}$.
- **Three-reservoir (branching) problem:** continuity at the junction J ($\Sigma Q = 0$) plus energy equation for every pipe. The piezometric head at J is guessed and corrected until the flows balance.

✎ **Exam Question:** A pipe 300 mm diameter, 500 m long, carries water at 0.12 m^3/s. Take f = 0.02. Find the head loss. *Ans: $V = Q/A = 1.70$ m/s; $h_f = 0.02 \times 500 \times 1.70^2/(2 \times 9.81 \times 0.3) \approx 4.9$ m.*

2. REYNOLDS TRANSPORT THEOREM (RTT) AND APPLICATIONS

2.1 System and control volume

A **system** is a fixed collection of fluid particles (Lagrangian view). A **control volume (CV)** is a fixed region in space through which fluid flows (Eulerian view). The conservation laws (mass, momentum, energy) are naturally stated for a system, but we usually measure quantities in a CV. The RTT converts one view into the other.

2.2 Statement of the theorem

Let B be any extensive property (mass, momentum, energy) and b = dB/dm the corresponding intensive property (per unit mass). Then

$$\mathbf{dB_{sys}/dt = \partial/\partial t \int_{CV} \rho\, b\, dV + \oint_{CS} \rho\, b\, (V\cdot n)\, dA}$$

- **Term 1** (left): time rate of change of B for the system.
- **Term 2:** rate of change of B stored inside the control volume (unsteady term).
- **Term 3:** net flux of B through the control surface (convective term). (**V**·**n**) is positive for outflow and negative for inflow.

★ **Key Point:** For steady flow the unsteady term is zero. For one-dimensional flow with uniform properties across inlet and outlet, the surface integral reduces to $(\rho bVA)_{out} - (\rho bVA)_{in}$.

2.3 Derivation (one-dimensional idea)

Consider a system that occupies the CV at time t and has moved slightly at time t + Δt. $B_{sys}(t) = B_{CV}(t)$. At t + Δt the system equals the CV contents, minus the part that has flowed out through the outlet (region II), plus the part that has flowed in (region I) through the inlet. Therefore

$dB_{sys}/dt = \lim [B_{CV}(t+\Delta t) - B_{CV}(t)]/\Delta t + (\text{B-flux out}) - (\text{B-flux in})$,

which on taking the limit Δt → 0 gives the RTT above.

2.4 Applications of RTT

(i) Conservation of mass: put B = m, b = 1.

$$\mathbf{0 = \partial/\partial t \int_{CV} \rho\, dV + \oint_{CS} \rho\, (V\cdot n)\, dA}$$

For steady incompressible flow: $\Sigma Q_{in} = \Sigma Q_{out}$. At a pipe junction this is the node law used in network analysis.

(ii) Linear momentum: put B = m**V**, b = **V**.

$$\mathbf{\Sigma F = \partial/\partial t \int_{CV} \rho V\, dV + \oint_{CS} \rho V\, (V\cdot n)\, dA}$$

Applications: force on a pipe bend, jet on a vane, thrust on a nozzle, and, importantly, the **pressure surge** relation in water hammer (Section 4).

(iii) Energy equation: put B = E, b = e = u + $V^2/2$ + gz.

$$\mathbf{\dot{Q} - \dot{W}_s = \partial/\partial t \int_{CV} \rho e\, dV + \oint_{CS} \rho(e + p/\rho)(V\cdot n)\, dA}$$

Reduces to Bernoulli's equation with head loss for steady flow: $p_1/\gamma + \alpha_1 V_1^2/2g + z_1 + h_{pump} = p_2/\gamma + \alpha_2 V_2^2/2g + z_2 + h_{turbine} + h_L$.

α is the **kinetic energy correction factor** (α = 2 for laminar pipe flow, ≈ 1.05 for turbulent).

(iv) Angular momentum: B = **r** × m**V**; used for turbines, pumps and sprinklers.

2.5 Worked example – force on a pipe bend

Water (Q = 0.2 m^3/s) flows through a horizontal 90° reducing bend from 300 mm to 200 mm diameter. Inlet pressure 200 kPa.

- $V_1 = 0.2/(\pi \cdot 0.15^2) = 2.83$ m/s; $V_2 = 0.2/(\pi \cdot 0.1^2) = 6.37$ m/s.
- Bernoulli (ignoring loss): $p_2 = 200\,000 + 500(2.83^2 - 6.37^2) = 200\,000 - 16\,280 \approx 183.7$ kPa.
- x-direction: $p_1A_1 - F_x = \rho Q(0 - V_1) \rightarrow F_x = 200\,000 \times 0.0707 + 1000 \times 0.2 \times 2.83 \approx 14.7$ kN.
- y-direction: $F_y - p_2A_2 = \rho QV_2 \rightarrow F_y = 183\,700 \times 0.0314 + 1000 \times 0.2 \times 6.37 \approx 7.04$ kN.
- Resultant $F = \sqrt{(14.7^2 + 7.04^2)} \approx 16.3$ kN. The anchor block must resist this force.

✎ **Exam Question:** State the Reynolds Transport Theorem. Deduce the continuity and momentum equations for a control volume. (10 marks)

3. STEADY FLOW ANALYSIS OF PIPE NETWORK SYSTEMS

3.1 What is a pipe network?

A pipe network is a set of pipes connected at **nodes (junctions)** forming one or more **closed loops**. Typical examples: municipal water-distribution systems, irrigation networks, fire-fighting grids. Reasons for loops: reliability (water can reach a node by more than one path) and balancing of pressure.

3.2 Basic laws for network analysis

Law 1 – Continuity at every node:

$$\Sigma Q_{in} - \Sigma Q_{out} = q_{node} \text{ (external demand/supply)}$$

Law 2 – Energy conservation around every closed loop:

$$\Sigma h_f = 0 \text{ (clockwise losses positive, anticlockwise negative)}$$

Law 3 – Head-discharge relation for each pipe:

$$h_f = r\,Q^n$$

where r is the pipe resistance and n is the flow exponent (n = 2 for Darcy–Weisbach with constant f, n = 1.852 for Hazen–Williams).

Formula	Resistance r	Exponent n
Darcy–Weisbach	$8fL/(\pi^2 g D^5)$	2
Hazen–Williams	$10.67\ L/(C^{1.852}\ D^{4.87})$	1.852
Manning	$10.29\ n^2 L/D^{5.33}$	2

3.3 Number of equations

If a network has P pipes, J junctions and L loops, then P = J + L − 1. We have (J − 1) independent node equations and L loop equations; with P unknown flows the system is exactly determined. The equations are non-linear (because of Q^n), so they must be solved iteratively.

3.4 Hardy Cross method (loop-balancing method)

Developed by Hardy Cross (1936). It assumes initial flows that satisfy continuity at every node, then corrects each loop by a flow ΔQ so that Σh_f approaches zero.

Let Q_0 be the assumed flow and ΔQ the correction. Then $Q = Q_0 + \Delta Q$ and

$h_f = r(Q_0 + \Delta Q)^n \approx r\,Q_0^{\,n} + n\,r\,Q_0^{\,n-1}\,\Delta Q$ (binomial expansion, higher terms dropped).

Imposing $\Sigma h_f = 0$ for the loop:

$$\Delta Q = -\,\Sigma\, r\,Q_0|Q_0|^{n-1} \,/\, \Sigma\, n\,r\,|Q_0|^{n-1}$$

For n = 2 and clockwise-positive convention: **$\Delta Q = -\,\Sigma\, rQ|Q| \,/\, (2\,\Sigma\, r|Q|)$**.

Step-by-step procedure

- Step 1: Draw the network, number pipes, nodes and loops; mark known inflows/outflows.
- Step 2: Assume a distribution of flows satisfying continuity at each node. Mark the direction (clockwise +ve, anticlockwise −ve).
- Step 3: Compute r for each pipe from D, L and friction formula.
- Step 4: For each loop compute $\Sigma\, rQ|Q|^{n-1}$ (head-loss sum) and $\Sigma\, n\,r|Q|^{n-1}$.

- Step 5: Compute ΔQ and apply it to all pipes in that loop. A pipe common to two loops receives ΔQ from both loops (with opposite sign for the neighbouring loop).
- Step 6: Repeat until |ΔQ| is smaller than the tolerance (e.g. 0.0001 m^3/s) or $\Sigma h_f \approx 0$.

★ **Key Point:** Hardy Cross does NOT require pressure/heads to be assumed – it works with flows only. Its convergence is first-order (linear) and can be slow for large networks, which is why Newton–Raphson methods are used in software.

3.5 Worked example – two-loop network

Consider a single loop ABCD (r in s^2/m^5, n = 2) with assumed flows (clockwise +ve): AB = +40 L/s, BC = +10 L/s, CD = −20 L/s (i.e. anticlockwise flow from D to C), DA = −30 L/s, and r = 400, 800, 300, 500 respectively.

Pipe	Q (m^3/s)	r	h = rQ	Q		2r	Q	
AB	+0.040	400	+0.640	32.0				
BC	+0.010	800	+0.080	16.0				
CD	−0.020	300	−0.120	12.0				
DA	−0.030	500	−0.450	30.0				

Σh = +0.150 m; Σ2r|Q| = 90.0. ΔQ = −0.150/90.0 = −0.00167 m^3/s (anticlockwise correction). New flows: AB = 0.03833, BC = 0.00833, CD = −0.02167, DA = −0.03167. Repeat; after 2–3 iterations Σh → 0.

3.6 Method of nodal heads (H-equations)

Unknowns are the heads H_j at the nodes. For each node the continuity equation is written with $Q_{ij} = ((H_i - H_j)/r_{ij})^{1/n}$. The resulting set of J non-linear equations is solved by **Newton–Raphson**. This is the method used in EPANET and WaterGEMS.

3.7 Method of linear theory (Wood & Charles) and Newton–Raphson for Q-equations

- **Linear theory:** write $h_f = r\,Q_0^{n-1} \cdot Q = r' Q$, i.e. h_f is linear in Q with $r' = r|Q_0|^{n-1}$. Solve the linear system simultaneously, average the new and old flows, and repeat.
- **Simultaneous loop corrections (Newton–Raphson):** solve for all ΔQ_{loop} at once; converges quadratically, but needs matrix inversion.

3.8 Comparison of methods

Method	Unknowns	Convergence	Remark
Hardy Cross (loop)	ΔQ per loop	Linear, slow	Hand calculation; needs initial feasible flows
Newton–Raphson (node heads)	Node heads	Quadratic	Basis of EPANET; robust
Linear theory	Q in all pipes	Fast	Needs no initial feasible flows
Gradient method (Todini)	Q and H together	Quadratic	Handles pumps, valves, PRVs

✎ **Exam Question:** Explain the Hardy Cross method for analysing a water-distribution network and derive the expression for the flow correction ΔQ. (10 marks)

4. UNSTEADY FLOW IN PIPES – WATER HAMMER

4.1 Definition

Water hammer (hydraulic transient, pressure surge) is the series of pressure waves set up in a pressurised pipeline when the flow velocity is changed rapidly, e.g. by sudden valve closure, pump trip or turbine load rejection. Compressibility of water and elasticity of pipe wall, usually neglected in steady flow, become essential.

Causes: valve closure/opening, pump start-up or failure, turbine load rejection, air release, column separation and collapse of voids.

Consequences: pressures of several times static pressure → pipe rupture, joint failure, buckling by sub-atmospheric pressure, vibrations, noise.

4.2 Physical description (valve closed suddenly at downstream end)

Consider a pipeline of length L from a reservoir to a valve, steady velocity V_0. The valve closes instantly at t = 0.

- **0 < t < L/a:** The layer of fluid next to the valve stops. Its kinetic energy converts to elastic strain energy: pressure rises by ΔH and a pressure wave of celerity a travels upstream. Fluid ahead of the wave still has velocity V_0; behind it V = 0.
- **t = L/a:** the wave reaches the reservoir. The whole pipe is at pressure H_0 + ΔH, with V = 0.
- **L/a < t < 2L/a:** The pipe is higher in pressure than reservoir, so fluid flows back toward the reservoir; pressure restores to H_0 as a wave travels downstream with V = $-V_0$.
- **t = 2L/a:** The wave reaches the valve with V = $-V_0$ and p = p_0.
- **2L/a < t < 3L/a:** The fluid at the valve cannot continue to move away; a negative pressure wave (−ΔH) travels upstream, bringing V to 0.
- **3L/a < t < 4L/a:** Pressure returns to normal by the wave moving toward the valve. At t = 4L/a the cycle repeats.

Period of oscillation = 4L/a. In real pipes friction damps the oscillations.

4.3 Joukowsky (Allievi) equation

Apply the RTT momentum equation to a control volume moving with the wave front. Fluid of density ρ approaches the front at relative speed (a + V_0), leaves at (a + V). Neglecting small terms:

$$\Delta H = -(a/g)\,\Delta V \quad \text{or} \quad \Delta p = -\rho\, a\, \Delta V$$

For complete stoppage ΔV = $-V_0$, so $\Delta H = a\,V_0/g$.

Example: a = 1000 m/s, V_0 = 2 m/s → ΔH = 1000 × 2/9.81 ≈ 204 m (≈ 2.0 MPa). This is why rapid valve closure is dangerous.

4.4 Wave speed (celerity)

$$a = \sqrt{(K/\rho)} \,/\, \sqrt{(1 + (K\,D / E\,e)\,c_1)}$$

- K = bulk modulus of the liquid; ρ = density; D = pipe diameter; e = wall thickness; E = Young's modulus of pipe material.
- c_1 is the pipe-support factor: for thin-walled pipe anchored at upstream end $c_1 = 1 - \mu_p/2$; anchored throughout against longitudinal movement $c_1 = 1 - \mu_p^2$ (μ_p = Poisson's ratio).
- For a rigid pipe (E → ∞): a = √(K/ρ) ≈ 1450 m/s for water.

Pipe material	Typical wave speed a (m/s)
Rigid (theoretical)	1450
Steel	1000 – 1250
Concrete / cast iron	1000 – 1200
Asbestos cement	900 – 1100
PVC / PE	200 – 500

4.5 Classification of valve closure

- **Critical period** T_c = 2L/a, the time taken for the pressure wave to travel to the reservoir and return.
- **Rapid (sudden) closure:** $T_{closure} \leq 2L/a$. The full Joukowsky rise occurs: $\Delta H = aV_0/g$.
- **Slow closure:** $T_{closure} > 2L/a$. Reflected relief wave arrives before closure is finished, so the maximum rise is smaller. Approximate (Michaud): $\Delta H \approx 2\ L\ V_0\ /(g\ T_{closure})$.

★ **Key Point:** Closing the valve in a time > 2L/a does not eliminate transients, but reduces them significantly. Surge tanks, air vessels, pressure-relief valves and flywheels are used to control them.

✎ **Exam Question:** A steel pipe, 1.2 m diameter, 8 mm thick, 1500 m long carries water at 2.5 m/s. Compute the pressure rise for a valve closure in 1.0 s. (Use a = 1000 m/s.) *Ans: 2L/a = 3 s; closure 1 s < 3 s → rapid. ΔH = 1000 × 2.5/9.81 ≈ 255 m.*

5. BASIC EQUATIONS OF WATER HAMMER

5.1 Assumptions

- Flow is one-dimensional; velocity V(x,t) and piezometric head H(x,t) are cross-section averages.
- Pipe is full, linear-elastic and prismatic; wave speed a is constant.
- Liquid is slightly compressible; density variation is small except where multiplied by a.
- Unsteady friction is represented by the steady-state Darcy–Weisbach term.
- Convective terms V∂V/∂x and V∂H/∂x are small compared with other terms because V ≪ a.

5.2 Continuity equation

Apply the mass conservation to an elementary pipe length δx. The mass accumulates due to (i) liquid compression and (ii) pipe-wall expansion, both functions of pressure change. This gives

$$\partial H/\partial t + (a^2/g)\,\partial V/\partial x + V\,\partial H/\partial x + V \sin\theta = 0 \rightarrow \partial H/\partial t + (a^2/g)\,\partial V/\partial x = 0 \text{ (simplified)}$$

Here θ is the pipe inclination.

5.3 Momentum (dynamic) equation

Apply Newton's second law to the fluid element of length δx: pressure gradient + gravity + wall shear = mass × acceleration.

$$\partial V/\partial t + g\,\partial H/\partial x + (f\,V|V|)/(2D) = 0$$

Here the last term is the wall friction (Darcy–Weisbach form, with |V| to retain direction).

5.4 Summary – the two governing PDEs

	Form		
	$\partial H/\partial t + (a^2/g)\,\partial V/\partial x = 0$		
	$\partial V/\partial t + g\,\partial H/\partial x + f\,V$	V	/(2D) = 0

They are a pair of coupled, quasi-linear, hyperbolic partial differential equations with two unknowns H(x,t) and V(x,t). Analytical solutions exist only for very idealised cases (frictionless, linear), so numerical solution – the **method of characteristics (MOC)** – is used.

5.5 Rigid-column theory vs elastic theory

- **Rigid-column theory:** neglects compressibility; the entire liquid column accelerates as one body. Governed by L/g dV/dt = ΔH. Valid for slow transients (gradual valve operation, surge tanks, mass oscillation).
- **Elastic (water hammer) theory:** includes compressibility and wall elasticity. Valid for fast transients where pressure waves travel ($T_{closure} \lesssim 10$ L/a).

6. SOLUTION BY THE METHOD OF CHARACTERISTICS (MOC)

6.1 Idea

MOC converts the two PDEs into four ordinary differential equations valid along special lines (characteristic lines) in the x–t plane. These ODEs are integrated by finite differences along the characteristics. It handles non-linear friction, complex boundary conditions (valves, pumps, reservoirs, surge tanks) and is the industry standard (e.g. Wylie & Streeter).

6.2 Derivation

Let $L_1 = \partial V/\partial t + g\,\partial H/\partial x + (f\,V|V|)/(2D) = 0$ (momentum)

Let $L_2 = (g/a^2)\,\partial H/\partial t + \partial V/\partial x = 0$ (continuity)

Form the linear combination $L = L_1 + \lambda\,L_2$:

$(\partial V/\partial t + \lambda\,\partial V/\partial x) + \lambda\,(g/a^2)\,(\partial H/\partial t + (a^2/\lambda)\,\partial H/\partial x) + f\,V|V|/(2D) = 0.$

Choose λ so that dx/dt is the same for V and H: $dx/dt = \lambda$ and $dx/dt = a^2/\lambda$, hence $\lambda = \pm a$. Then the bracketed terms become total derivatives dV/dt and dH/dt:

$$\mathbf{C^+:\ (g/a)\,dH/dt + dV/dt + f\,V|V|/(2D) = 0 \text{ valid along } dx/dt = +a}$$

$$\mathbf{C^-:\ -(g/a)\,dH/dt + dV/dt + f\,V|V|/(2D) = 0 \text{ valid along } dx/dt = -a}$$

6.3 Finite-difference form

Divide the pipe into N equal reaches $\Delta x = L/N$ and choose $\Delta t = \Delta x/a$ (Courant condition, so characteristics pass exactly through grid points). To find conditions at point P (position i, time t + Δt) use points A (i−1, t) and B (i+1, t):

$$\mathbf{C^+:\ H_P = C_P - B_P\,Q_P \text{ where } C_P = H_A + B\,Q_A - R\,Q_A|Q_A|}$$

$$\mathbf{C^-:\ H_P = C_M + B_M\,Q_P \text{ where } C_M = H_B - B\,Q_B + R\,Q_B|Q_B|}$$

Here $B = a/(gA)$ is the **pipe characteristic impedance**, $R = f\,\Delta x/(2\,g\,D\,A^2)$ is the **resistance coefficient**, and Q is discharge. Solve the two equations for the two unknowns (Q_P, H_P) at interior nodes:

$$\mathbf{Q_P = (C_P - C_M)/(B_P + B_M),\ H_P = (C_M\,B_P + C_P\,B_M)/(B_P + B_M)}$$

6.4 Boundary conditions

- **Upstream reservoir (constant head H_R):** $H_P = H_R$; only C^- is available → $Q_P = (H_R - C_M)/B_M$.
- **Downstream dead end / closed valve:** $Q_P = 0$; only C^+ → $H_P = C_P$.
- **Downstream valve (orifice):** $Q_P = (C_d A_G)_P\,\sqrt{(2g\,H_P)}$; combine with C^+ to get a quadratic in Q_P.
- **Series junction (change of diameter):** continuity $Q_{P1} = Q_{P2}$ and equal head (neglecting losses); two C^+, C^- equations.
- **Branching junction:** $\Sigma Q = 0$, equal head H at the node; N + 1 unknowns with N characteristics.
- **Pump:** head–discharge curve $H_{pump} = a_0 + a_1 Q + a_2 Q^2$ combined with the characteristic.
- **Surge tank:** continuity of tank level z_s + equal heads.

6.5 Computational steps

- 1. Compute steady-state initial values of H and Q at all nodes (t = 0).

- 2. Select N, $\Delta x = L/N$, $\Delta t = \Delta x/a$.
- 3. For each time step, compute interior nodes using C^+ and C^-.
- 4. Apply the boundary conditions at the two ends.
- 5. Store H and Q; advance $t = t + \Delta t$ until the required duration is reached.
- 6. Plot H versus t at critical points (valve, mid-point) and envelope of maximum/minimum head along the pipe.

★ **Key Point:** If pipes have different wave speeds, the grid spacing is adjusted so that Δt is common to all pipes ($\Delta x_i = a_i \Delta t$), or $\Delta t = \Delta x/a$ is slightly altered ("wave-speed adjustment" interpolation).

6.6 Worked example – frictionless line

Reservoir head 100 m, L = 600 m, a = 1200 m/s, V_0 = 1.5 m/s, valve closed instantly. N = 2, Δx = 300 m, Δt = 0.25 s. At t = 0.25 s: valve node $H = 100 + a V_0/g = 100 + 183.5 = 283.5$ m; at middle node H = 100 m; V = 0 at valve. At 0.5 s: wave has reached reservoir; the middle node reaches 283.5 m. At 1.0 s (= 2L/a) the pipe is at rest, high pressure. At t = 1.0–2.0 s the wave reverses. A plot of the valve head vs. time is a **square wave** between 283.5 m and −83.5 m (limited to vapour pressure, causing column separation) of period 4L/a = 2.0 s.

✎ **Exam Question:** Derive the characteristic equations of water hammer and explain the finite-difference solution procedure. (15 marks)

7. WATER HAMMER CONTROL AND PROTECTION

7.1 Methods of controlling surges

- **Slow valve operation:** keep closure time T_c > 2L/a, preferably > 5 L/a.
- **Surge tank (surge shaft):** an open standpipe connected near the turbine or valve. It reflects the pressure wave and provides/absorbs water. Types: simple, restricted orifice, differential (Johnson), closed (air) surge tank.
- **Air vessel (hydropneumatic accumulator):** a closed vessel with compressed air cushion; effective at pumping stations.
- **One-way surge tank / feed tank:** supplies water only when the pipe pressure drops below tank level to prevent column separation.
- **Pressure-relief / surge-anticipation valves:** open at preset pressure and release flow.
- **Air-inlet and vacuum-breaker valves** at high points to avoid sub-atmospheric pressure and pipe collapse.
- **Flywheel on pump:** increases inertia so pump coasts down slowly after power failure.
- **Bypass line with non-return valve** around the pump.

7.2 Column separation

When pressure falls to the vapour pressure of water, vapour cavities form and the liquid column separates. Subsequent collapse of the cavity causes a sharp pressure spike that can exceed the Joukowsky value. It is modelled by the **discrete vapour cavity model (DVCM)** within MOC.

7.3 Pipe design for surge

Pipe pressure class must exceed (static head + surge head). Required wall thickness from the hoop-stress equation: $e = p\,D / (2\,\sigma_{allow})$. The wave speed a decreases with decreasing wall thickness, so thin pipes have smaller surge pressures but must still resist the total pressure.

8. NETWORK ANALYSIS – MODERN COMPUTER APPROACH

8.1 Steady-state analysis in software (EPANET)

Computer programs treat the network as a graph (links = pipes, pumps, valves; nodes = junctions, reservoirs, tanks). Input: pipe lengths, diameters, roughness, node demands, elevations, pump curves. Output: flows, velocities, pressures at nodes.

Gradient algorithm: solves the energy equation for all links and the continuity equation for all nodes simultaneously through Newton–Raphson iterations, linearised each time:

$$\mathbf{A_{11}\,Q + A_{12}\,H = -A_{10}\,H_0,\ A_{21}\,Q = q}$$

where A_{12} is the connectivity matrix (incidence), A_{11} is a diagonal matrix of head-loss gradients n r$|Q|^{n-1}$, and q is the demand vector.

8.2 Extended-period simulation (EPS)

Demand patterns, tank filling/draining, pump on–off controls simulated through time using successive steady-state analyses. Useful for planning storage and pumping schedules.

8.3 Design criteria for distribution networks

- Minimum residual pressure at nodes: 7–15 m for ordinary domestic supply (≥ 17 m for 3-storey building).
- Maximum velocity ≈ 1.5–2.0 m/s; minimum ≈ 0.3 m/s to avoid sedimentation.
- Loops of equal head-loss; no dead-ends where possible.
- Fire flow demand (e.g. 100 L/s for 4 hr) to be checked.

8.4 Transient network analysis

Surge analysis of branched/looped networks uses the MOC at every pipe, with junction compatibility equations at nodes. Software: HAMMER, SURGE, ALLIEVI, TSNet. Steps: (a) steady state initialisation, (b) event definition (pump trip, valve closure), (c) time-marching MOC, (d) envelope of max/min heads, (e) design of protection devices, (f) re-simulation.

8.5 Equivalent pipe concept and Dupuit equation

Two pipes are equivalent if the same discharge gives the same head loss. For a series system: $L_e/D_e^{4.87} = \Sigma\, L_i/D_i^{4.87}$ (H–W, with same C). For Dupuit equation in the parallel case: $Q = \Sigma\, Q_i$, and with $D_e^{2.63}/\sqrt{L_e^{0.54}} = \Sigma\, D_i^{2.63}/L_i^{0.54}$.

8.6 Quick revision – formulae sheet

	Q	$^{n-1}$ / Σ n r	Q

8.7 Important short questions

- Define water hammer. What are its causes?

- State the Joukowsky equation and give its limitation.
- Why is the Courant condition $\Delta t = \Delta x/a$ important in MOC?
- Distinguish between rigid column theory and elastic theory.
- List four methods of controlling surge pressure.
- Why is Hardy Cross method called a loop method? Write the condition at each node and loop.